

Navigation icons: back, forward, search, etc.

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Took 0 sec. Last updated by anonymous at July 23 2025, 5:44:05 PM. (outdated)

```
-- Drop existing table if an then create an external table 'wdi_csv_text'
-- Data is CSV format with comma delimiters, stored at an HDFS location.

DROP TABLE IF EXISTS wdi_csv_text;
CREATE EXTERNAL TABLE wdi_csv_text(
  year INTEGER, countryName STRING, countryCode STRING, indicatorName STRING, indicatorCode STRING, indicatorValue FLOAT)
ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' LINES TERMINATED BY '\n'
LOCATION 'hdfs://user/adityakjanch107/nive/wdi_csv_text';
```

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Took 15 sec. Last updated by anonymous at July 17 2025, 5:07:58 PM. (outdated)

```
-- Overwrite data in wdi_csv_text with all records from wdi_gs table
INSERT OVERWRITE TABLE wdi_csv_text
SELECT * FROM wdi_gs
```


HIVE JOB FINISHED

Took 2 min 19 sec. Last updated by anonymous at July 17 2025, 5:11:12 PM. (outdated)

```
-- Run twice to observe filesystem caching effects
SELECT Count(countryname) FROM wdi_csv_text;
```









[settings](#)

Took 37 sec. Last updated by anonymous at July 21 2025, 2:33:32 PM. (outdated)

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First run took 28.502 seconds to generate the output

Second run took **22.010 seconds** for the same query.

Took 0 sec. Last updated by anonymous at July 23 2025, 5:44:12 PM.

```
-- Clear filesystem cache and re-run count query to compare performance
SELECT count(countryName) FROM wdi_csv_text
```


HIVE JOB FINISHED

Took 36 sec. Last updated by anonymous at July 17 2025, 5:45:37 PM. (outdated)

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It took **36 seconds** for the system to run the same query.

Before the third run, I cleared the filesystem cache using the below command to see if there's any difference.

Took 0 sec. Last updated by anonymous at July 23 2025, 5:44:17 PM. (outdated)

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Took 0 sec. Last updated by anonymous at July 23 2025, 5:44:19 PM.

```
%ish
cd ~
hdfs dfs -get hdfs:///user/adityakhanjani07/hive/wdi/wdi_csv_text
cd wdi_csv_text
du -ch
echo 3 | sudo tee /proc/sys/vm/drop_caches
date +%s %s cat * | wc %s date +%s
```

FINISHED

Took 15 sec. Last updated by anonymous at July 17 2025, 6:22:08 PM. (outdated)

TASK: Count the rows in the dataset.

HIVE: 36 seconds
BASH: 17 seconds

The delta is of a whopping 20 seconds.

This is primarily due to their architectural design.

FINISHED

basn operates directly on the niesystem witrn minimai ovearoad, making it quick for smail and locai tasks.

Hive, on the other hand goes through multiple layers to process the data. It involves parsing SQL, accessing metadata from metastore, applying schema checks and later triggering a MR | Tez job.

While Hive was slow for this task, it is optimal for a larger and complex data job.

Took 0 sec. Last updated by anonymous at July 23 2025, 5:44:22 PM.

Parsing Issues

Took 0 sec. Last updated by anonymous at July 23 2025, 5:44:28 PM.

```
-- List distinct indicator codes, ordered alphabetically, limited to 20 results
SELECT distinct(indicatorcode)
FROM wdi_csv_text
ORDER BY indicatorcode
LIMIT 20;

-- Observe the values under indicatorCode column
```

Took 1 sec. Last updated by anonymous at July 17 2025, 7:29:17 PM. (outdated)

The above ouput indicates that there could be parsing issues. The next steps include diagnosing and debugging by creating a debug table.

Took 0 sec. Last updated by anonymous at July 23 2025, 5:44:30 PM. (outdated)

```
--Creating a new debug table with a single column
DROP TABLE IF EXISTS wdi_gs_debug;
CREATE EXTERNAL TABLE wdi_gs_debug (
  Column1 STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\b'
LINES TERMINATED BY '\n'
LOCATION 'gs://jarvis_data_eng_editya/datasets/wdi_2016';
```

Took 1 sec. Last updated by anonymous at July 17 2025, 7:47:17 PM. (outdated)

```
| SELECT * FROM wdi_gs_debug LIMIT 10;
```

settings

column1
1960,Chad,TCD,Income share held by second 20%,SI.DST.02ND.20,0
1960,Chad,TCD,"Net enrolment rate, primary, both sexes (%)",SE.PRM.NENR,0
1960,Chad,TCD,"Lower secondary completion rate, total (% of relevant age group)",SE.SEC.CMPT.LO.ZS,0
1960,Chad,TCD,Mothers are guaranteed an equivalent position after maternity leave (1=yes; 0=no),SG.MMR.LEVE.EP,0
1960,Chad,TCD,"Currency composition of PPG debt, French franc (%)",DT.CUR.FFRFC.ZS,0
1960,Chad,TCD,"Annual freshwater withdrawals, industry (% of total freshwater withdrawal)",ER.H2O.FWIN.ZS,0
1960,Chad,TCD,Gross capital formation (% of GDP),NE.GDI.TOTL.ZS,13.5736311608589
1960,Chad,TCD,"PPG, other private creditors (DOD, current US\$)",DT.DOD.PROP.CD,0

Took 1 sec. Last updated by anonymous at July 17 2025, 7:47:27 PM. (outdated)

```
-- Check rows containing "%( % of urban population)%" to verify parsing of indicatorCode values
SELECT * FROM wdi_gs_debug
WHERE column1 LIKE "%(% of urban population)%";
```

settings

column1
1960,Argentina,ARG,"Access to non-solid fuel, urban (% of urban population)",EG.NSF.ACCS.UR.ZS,0
1960,American Samoa,ASM,"Access to electricity, urban (% of urban population)",EG.ELC.ACCS.UR.ZS,0
1960,Belgium,BEL,"People practicing open defecation, urban (% of urban population)",SH.STA.ODFC.UR.ZS,0
1960,Bangladesh,BGD,"Access to non-solid fuel, urban (% of urban population)",EG.NSF.ACCS.UR.ZS,0
1960,Bangladesh,BGD,Population in the largest city (% of urban population),EN.URB.LCTY.UR.ZS,20.521181719086
1960,Bosnia and Herzegovina,BIH,Population in the largest city (% of urban population),EN.URB.LCTY.UR.ZS,27.664469115833
1960,Bermuda,BMU,Urban poverty headcount ratio at national poverty lines (% of urban population),SI.POV.URHC,0
1960,Barbados,BRB,"People practicing open defecation, urban (% of urban population)",SH.STA.ODFC.UR.ZS,0

Took 47 sec. Last updated by anonymous at July 17 2025, 8:22:08 PM. (outdated)

```
-- Create a Table with OpenCSV SerDe
DROP TABLE IF EXISTS wdi_opencsv_gs;
CREATE EXTERNAL TABLE wdi_opencsv_gs (
  year INT,
  countryName STRING,
  countryCode STRING,
  indicatorName STRING,
  indicatorCode STRING,
  indicatorValue FLOAT
)
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
WITH SERDEPROPERTIES (
  "separatorChar" = ",",
  "quoteChar" = "\""
)
STORED AS TEXTFILE
LOCATION 'gs://jarvis_data_eng_editya/datasets/wdi_2016'
TBLPROPERTIES (
  "skip.header.line.count" = "1"
)
```

Took 0 sec. Last updated by anonymous at July 17 2025, 7:56:16 PM. (outdated)

```
-- Verify if `wdi_opencsv_gs` table correctly parsed `indicatorCode` values using OpenCSVSerde
SELECT * FROM wdi_opencsv_gs LIMIT 10;

-- Observe the values under `indicatorcode` column
```

settings

year	countryname	countrycode	indicatorname	indicatorcode	indicatorvalue
1960	Chad	TCD	Community health workers (per 1,000 people)	SH.MED.CMH.W.P3	0
1960	Chad	TCD	Income share held by second 20%	SI.DST.02ND.20	0
1960	Chad	TCD	Net enrolment rate, primary, both sexes (%)	SE.PRM.NENR	0
1960	Chad	TCD	Lower secondary completion rate, total (% of relevant age group)	SE.SEC.CMPT.LO.ZS	0
1960	Chad	TCD	Mothers are guaranteed an equivalent position after maternity leave (1=yes; 0=no)	SG.MMR.LEVE.EP	0
1960	Chad	TCD	Currency composition of PPG debt, French	DT.CUR.FFRFC.ZS	0

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Took 1 sec. Last updated by anonymous at July 17 2025, 8:30:50 PM. (outdated)

Took 2 min 23 sec. Last updated by anonymous at July 21 2025, 2:58:51 PM. (outdated)

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Took 1 sec. Last updated by anonymous at July 21 2025, 2:59:03 PM. (outdated)

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Took 37 sec. Last updated by anonymous at July 17 2025, 8:30:15 PM. (outdated)

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Took 0 sec. Last updated by anonymous at July 23 2025, 5:44:40 PM.

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Took 0 sec. Last updated by anonymous at July 23 2025, 5:44:41 PM. (outdated)

HIVE JOB FINISHED

Took 1 min 29 sec. Last updated by anonymous at July 21 2025, 3:01:56 PM.



HIVE JOB FINISHED

Took 35 sec. Last updated by anonymous at July 21 2025, 3:36:25 PM. (outdated)

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It took `wdi_opencsv_text` 1m 19s 55ms to complete the query.
It took `wdi_csv_text` 23s 210ms for the same operation.

There is a performance delta of 55s 340ms, where wdi_csv_text performed significantly faster. This suggests that using a simpler SerDe (like DELIMITED) may be more efficient in certain contexts where complex parsing is unnecessary.

Took 0 sec. Last updated by anonymous at July 23 2025, 9:44:47 PM.

OpenCSVSerde limitaion

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Took 0 sec. Last updated by anonymous at July 23 2025, 9:44:51 PM.

-- View detailed metadata and storage information for the table 'wdi_opencsv_gs'

FINISHED

DESCRIBE FORMATTED wdi_opencsv_gs;

settings

col_name	data_type	comment
# col_name	data_type	comment
year	string	from deserializer
countryname	string	from deserializer
countrycode	string	from deserializer
indicatorname	string	from deserializer
indicatorcode	string	from deserializer
indicatorvalue	string	from deserializer
	null	null

Took 0 sec. Last updated by anonymous at July 21 2025, 2:37:22 PM. (outdated)

-- Drop the view if it exists, then create a new view to simplify querying 'wdi_opencsv_gs'

FINISHED

DROP VIEW IF EXISTS wdi_opencsv_text_view;
CREATE VIEW IF NOT EXISTS wdi_opencsv_text_view
AS
SELECT * FROM wdi_opencsv_gs;

Took 0 sec. Last updated by anonymous at July 21 2025, 3:34:40 PM. (outdated)

2015 Canada GDP Growth HQL

FINISHED

Took 0 sec. Last updated by anonymous at July 23 2025, 9:44:54 PM.

-- HQL to find out the correct indicator name/code

HIVE JOB FINISHED

SELECT DISTINCT indicatorName, indicatorCode
FROM wdi_opencsv_text
WHERE indicatorName LIKE '%GDP growth (annual %)%';

settings

indicatorname	indicatorcode
GDP growth (annual %)	NY.GDP.MKTP.KD.ZG

Took 1 min 47 sec. Last updated by anonymous at July 21 2025, 4:41:46 PM. (outdated)

-- HQL to find 2015 Canada GDP growth

HIVE JOB FINISHED

SELECT indicatorvalue AS GDP_growth_value, year as Year, countryName
FROM wdi_opencsv_text
WHERE countryName = 'Canada'
AND year = 2015
AND IndicatorCode = 'NY.GDP.MKTP.KD.ZG';

settings

gdp_growth_value	year	countryname
1.07826875075381	2015	Canada

Took 1 min 47 sec. Last updated by anonymous at July 21 2025, 4:44:58 PM. (outdated)

Why did the HQL query took a significant time to process?

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Since there were no partitions made on the table, Hive has to scan the entire dataset.

Took 0 sec. Last updated by anonymous at July 23 2025, 9:44:58 PM.

Hive Partitions

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Took 0 sec. Last updated by anonymous at July 23 2025, 9:45:02 PM.

-- Create wdi_opencsv_text_partitions table which partitioned by year

FINISHED

DROP TABLE IF EXISTS wdi_opencsv_text_partitions;
CREATE EXTERNAL TABLE wdi_opencsv_text_partitions (
countryName STRING,
countryCode STRING,
indicatorName STRING,
indicatorCode STRING,
indicatorValue FLOAT
)
PARTITIONED BY (year INT)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\n'
STORED AS TEXTFILE
LOCATION 'hdfs:///user/adityakhajanchi87/hive/wdi/wdi_opencsv_text_partitions'

Took 3 sec. Last updated by anonymous at July 21 2025, 10:06:27 PM. (outdated)

```
-- Load data from wdi_opencsv_text to wdi_opencsv_text_partitions using dynamic partitions
SET hive.exec.dynamic.partition = true;
SET hive.exec.dynamic.partition.mode = nonstrict;
INSERT OVERWRITE TABLE wdi_opencsv_text_partitions
PARTITION (year)
SELECT
  countryName,
  countryCode,
  indicatorName,
  indicatorCode,
  indicatorValue,
  year
FROM wdi_opencsv_text;
```

HIVE JOB FINISHED

Took 2 min 50 sec. Last updated by anonymous at July 21 2025, 10:10:52 PM. (outdated)

```
%sh
# Observe the number of partitions created for wdi_opencsv_text_partitions table

hdfs dfs -ls hdfs:///user/adiyakhajanchi07/hive/wdi/wdi_opencsv_text_partitions | grep "year=" | wc -l
```

FINISHED

Took 5 sec. Last updated by anonymous at July 21 2025, 10:07:47 PM. (outdated)

```
-- Re-run 2015 GDP Growth HQL against the wdi_opencsv_text_partitions table and compare execution time the previous solution
SELECT indicatorValue AS GDP_growth_value, year as Year, countryName
FROM wdi_opencsv_text_partitions
WHERE countryName = 'Canada'
AND year = 2015
AND indicatorCode = 'NY.GDP.MKTP.KD.ZG';
```

FINISHED

settings

gdp_growth_value	year	countryname
1.0762688	2015	Canada

Took 1 sec. Last updated by anonymous at July 21 2025, 10:18:24 PM. (outdated)

From 1m 35s 866ms to 2s !!
Partitioning significantly reduced query execution time.

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Took 0 sec. Last updated by anonymous at July 23 2025, 5:45:08 PM.

Columnar File Optimization

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Took 0 sec. Last updated by anonymous at July 23 2025, 5:45:09 PM.

```
-- Create a Parquet-backed table at the an HDFS location
DROP TABLE IF EXISTS wdi_csv_parquet;
CREATE EXTERNAL TABLE IF NOT EXISTS wdi_csv_parquet (
  year INT,
  countryName STRING,
  countryCode STRING,
  indicatorName STRING,
  indicatorCode STRING,
  indicatorValue FLOAT
)
STORED AS PARQUET
LOCATION 'hdfs:///user/adiyakhajanchi07/hive/wdi/wdi_csv_parquet';
```

FINISHED

Took 0 sec. Last updated by anonymous at July 21 2025, 10:26:57 PM. (outdated)

```
-- Load data from wdi_opencsv_gs to wdi_csv_parquet
INSERT OVERWRITE TABLE wdi_csv_parquet
SELECT *
FROM wdi_opencsv_gs;
```

HIVE JOB FINISHED

Took 2 min 30 sec. Last updated by anonymous at July 21 2025, 10:30:00 PM. (outdated)

```
%sh
#Compare file sizes between wdi_csv_parquet and wdi_opencsv_text

hdfs dfs -du -h /user/adiyakhajanchi07/hive/wdi
```

FINISHED

Took 5 sec. Last updated by anonymous at July 21 2025, 10:31:48 PM. (outdated)

Table Storage Size Comparison

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Table Name	Compressed Size	Uncompressed Size	HDFS Path
wdi_csv_parquet	131.9 MB	263.8 MB	/user/adiyakhajanchi07/hive/wdi/wdi_csv_parquet
wdi_csv_text	1.7 GB	3.4 GB	/user/adiyakhajanchi07/hive/wdi/wdi_csv_text
wdi_gs	0	0	/user/adiyakhajanchi07/hive/wdi/wdi_gs
wdi_opencsv_text	2.3 GB	4.5 GB	/user/adiyakhajanchi07/hive/wdi/wdi_opencsv_text
wdi_opencsv_text_partitions	1.9 GB	3.8 GB	/user/adiyakhajanchi07/hive/wdi/wdi_opencsv_text_partitions

Took 0 sec. Last updated by anonymous at July 23 2025, 5:45:15 PM.

Compare query execution times for counting records in two tables to evaluate performance differences

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Took 0 sec. Last updated by anonymous at July 23 2025, 5:45:17 PM.

```
-- Observe the query processing time to check total number of records in the `wdi_csv_parquet` table
SELECT Count(countryName) FROM wdi_csv_parquet;
```

HIVE JOB FINISHED

settings

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217504NR

Took 38 sec. Last updated by anonymous at July 21 2025, 10:42:42 PM. (outdated)

```
-- Observe the query processing time to check total number of records in the `wdi_opencsv_text` table
SELECT count(countryName) FROM wdi_opencsv_text;
```









[settings](#)

Took 1 min 38 sec. Last updated by anonymous at July 21 2025, 10:44:24 PM. (outdated)

Counting rows in `wdi_csv_parquet` took 21 seconds, showcasing faster query performance due to Parquet's columnar storage and compression

Counting rows in `wdi_opencsv_text` took 1 minute 22 seconds, indicating slower performance caused by raw CSV format and lack of optimizations

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Took 0 sec. Last updated by anonymous at July 23 2025, 5:45:21 PM.

```
-- Find the highest GDP growth value by year for each country
-- Output: GDP_growth_value, year, countryName

SELECT indicatorValue AS GDP_growth_value, year AS Year, countryName
FROM wdi_csv_parquet
WHERE countryName = 'Canada'
AND year = 2015
AND indicatorCode = 'NY.GDP.MKTP.KD.ZG';
```


HIVE JOB FINISHED

gdp_growth_value	year	countryname
1.0782688	2015	Canada

Took 39 sec. Last updated by anonymous at July 21 2025, 10:50:11 PM. (outdated)

```
-- Find the highest GDP growth value by year for each country
-- Output: GDP_growth_value, year, countryName

SELECT indicatorValue AS GDP_growth_value, year as Year, countryName
FROM wdi_opencsv_text
WHERE countryName = 'Canada'
AND year = 2015
AND indicatorCode = 'NY.GDP.MKTP.KD.ZG';
```


HIVE JOB FINISHED

gdp_growth_value	year	countryname
1.07826875075381	2015	Canada

Took 1 min 38 sec. Last updated by anonymous at July 21 2025, 10:52:10 PM. (outdated)

Query execution on `wdi_csv_parquet` is much faster (22s) than on `wdi_opencsv_text` (1m 22s), demonstrating the efficiency of Parquet's columnar storage and compression over CSV format.

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Took 0 sec. Last updated by anonymous at July 23 2025, 5:45:26 PM. (outdated)

Highest GDP Growth

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Took 0 sec. Last updated by anonymous at July 23 2025, 5:45:28 PM.

```
-- Highest GDP growth (NY.GDP.MKTP.KD.ZG) year for each country

SELECT countryName, countryCode, year, indicatorValue
FROM (
    SELECT
        countryName,
        countryCode,
        year,
        indicatorValue,
        ROW_NUMBER() OVER (PARTITION BY countryCode ORDER BY indicatorValue DESC) as rank
    FROM wds_csv_parquet
    WHERE indicatorCode = 'NY.GDP.MKTP.KD.ZG'
) ranked
WHERE rank = 1;
```



HIVE JOB FINISHED


[settings](#)

Took 43 sec. Last updated by anonymous at July 21 2025, 10:48:22 PM. (outdated)

```
FROM (
  SELECT
    countryName,
    countryCode,
    year,
    indicatorValue,
    ROW_NUMBER() OVER (PARTITION BY countryCode ORDER BY indicatorValue DESC) as rank
  FROM wdi_csv.parquet
  WHERE indicatorCode = 'NY.GOP.MKTR.KD.ZG'
) ranked
WHERE rank = 1;
```

Took 3 min 11 sec. Last updated by anonymous at July 21 2025, 10:57:55 PM. (outdated)

Hive execution time: 37 seconds, demonstrating efficient processing on Parquet with Hive.

SparkSQL execution time: 2 minutes 27 seconds, slower than **Hive** for this specific analytical query on the same dataset.

Took 0 sec. Last updated by anonymous at July 23 2025, 5:43:41 PM.

Took 0 sec. Last updated by anonymous at July 23 2025, 5:43:43 PM.

```
SELECT countryName, year, indicatorName, indicatorValue
FROM wdi_csv_parquet
WHERE indicatorCode = 'NY.GDP.MKTP.KD.ZG'
ORDER BY countryName, year;
```

Took 38 sec. Last updated by anonymous at July 21 2025, 11:16:06 PM. (outdated)

Took 0 sec. Last updated by anonymous at July 23 2025, 5:45:36 PM.